

IoT SMART METER With Dashboard

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Energy meters in homes track electricity usage, enabling accurate billing by governments and providers. However, traditional meters often rely on manual readings, prompting consumers and industries to seek smarter alternatives with real-time energy usage and billing updates. While smart meters fulfil these demands, they frequently involve invasive installations, requiring complex wiring changes that risk short circuits and discourage adoption. Additionally, they are typically costly.

Addressing these issues, a

non-invasive energy meter with an internet of things (IoT) dashboard capabilities has been introduced. This smart meter seamlessly attaches to devices, home wiring, and office or factory setups without altering existing wiring. It utilises an SCT013 non-invasive current

sensor and integrates a built-in Wi-Fi module, providing real-time updates on electricity consumption and billing.

Features of the IoT smart meter

This IoT smart meter offers several unique features:

BILL OF MATERIALS

Components	Description	Quantity
IndusBoard coin (U2)	3cm sized development board	1
SCT 013 000 100A (U1)	Non-invasive AC current sensor	1
OLED/ LCD display	SSD1306 / GC9A01	1
Resistors (R1, R2)	100k Ω	2
Resistor (R3)	33 Ω	1
Electrolytic capacitor (C1)	10 μ F, 25V	1

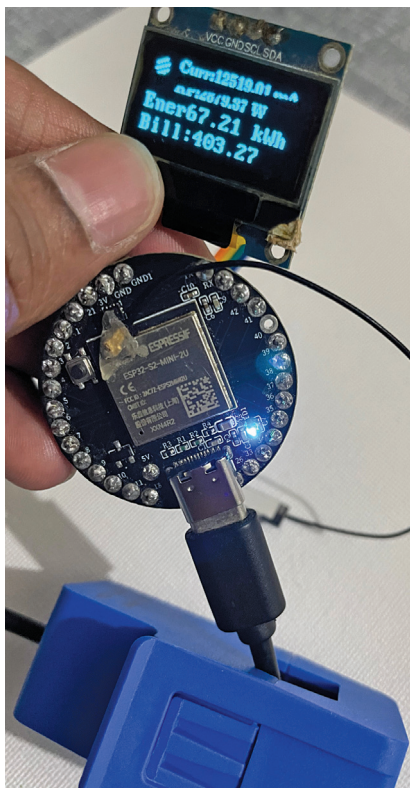


Fig. 1: Author's prototype of the IoT smart meter with SSD1306 OLED display

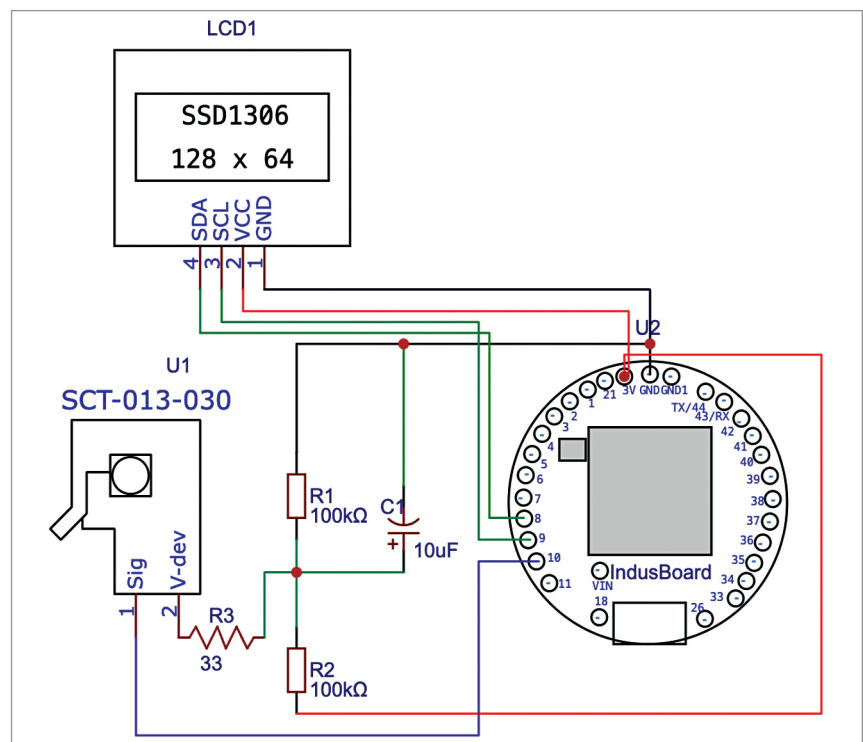


Fig. 2: Circuit diagram of the IoT smart meter with SSD1306 OLED display